Navin R

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Professional Summary

Information Technology student specializing in backend systems, distributed processing, and concurrency using Java and Spring Boot. Built fault-tolerant applications with retry pipelines, recovery services, and scalable asynchronous workflows.

Technical Skills

Languages: Java, Python, C++, SQL, Solidity
Frameworks: Spring Boot, React
Developer Tools: Git, Docker, Docker Compose, Maven
Databases: MySQL, MariaDB, PostgreSQL, Redis
Monitoring: Prometheus, Grafana, Micrometer
Operating System: Linux (Arch, Debian)
Core CS: Data Structures & Algorithms, OOP, Distributed Systems, Concurrency & Multi-threading, Blockchain Fundamentals

Education

Sri Krishna College of Technology — Coimbatore, TN
*Bachelor of Technology in Information Technology — **GPA: 7.78/10*** — *Expected Sep. 2027*

Projects

Mass Campaign Manager | *Spring Boot, React, PostgreSQL, Docker* — May 2026 – Present

- Architected a distributed campaign orchestrator handling 1M+ recipient workloads by delegating execution to a custom-built, decoupled worker microservice (Distributed Job Processor)
- Engineered a high-throughput data ingestion pipeline using chunked CSV streaming and 5K-batched PostgreSQL inserts to prevent JVM memory exhaustion during massive uploads
- Applied pessimistic write locking to safely serialize 200+ concurrent webhook callbacks, guaranteeing accurate cumulative execution states without data races
- Eliminated HTTP thread blocking via asynchronous CompletableFuture background processing, allowing the API to return 202 Accepted in milliseconds
- Designed idempotent failure recovery to re-process only incomplete chunks without duplication, monitored via a real-time React telemetry dashboard

Distributed Job Processor | *Spring Boot, Redis, PostgreSQL, Docker, Grafana* — Mar. – Apr. 2026

- Engineered a fault-tolerant distributed task queue processing jobs across multiple worker containers with at-least-once delivery guarantees
- Implemented atomic job claiming via Redis BLMOVE and distributed locks to prevent duplicate execution across concurrent workers
- Built an exponential backoff retry pipeline (10s→20s→40s) with a dead-letter queue, alongside Reaper and Reconciliation services for automated crash recovery
- Containerized the multi-node architecture via Docker Compose, instrumenting 13 custom Micrometer metrics to Prometheus and Grafana for live queue depth and throughput telemetry

Achievements

- Engineered a standalone, reusable distributed worker node architecture, fully documented and MIT-licensed for public use
- Solved 190+ algorithmic challenges on LeetCode, specializing in dynamic programming, graph theory, and advanced data structures
- Completed Google Git & GitHub Professional Certificate on Coursera
- Earned Java (Basic) certification from HackerRank